Quiz 1 - DSC 10, Fall 2026

Full Name:

TSN:

Quiz Time:

◯ 3:00	◯ 3:30	◯ 4:00	◯ 4:30
◯ 5:00	◯ 5:30	◯ 6:00	◯ 6:30
◯ 7:00	◯ 7:30	◯ 8:00	◯ 8:30

Instructions:

- This quiz consists of 5 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.
- Assume we have already run `import babypandas as bpd` and `import numpy as np`.

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

Coleman decides that tutoring DSC 10 is not lucrative enough to support his extravagant lifestyle, so he decides to open up a sushi restaurant called Coleman's Catch.

The DataFrame `menu` has a row for each menu item available at Coleman's Catch. A few rows are shown at right, but there are additional rows not shown. The `"Name"` of the menu item is in the index (string), and the columns are `"Type"` (string), `"Price"` in dollars (float), number of `"Pieces"` (int) per order, and average `"Rating"` from 0 to 5 (float).

Name	Type	Price	Pieces	Rating
Salmon Nigiri	Nigiri	6.5	2	4.7
Avocado Roll	Roll	7.0	8	4.2
Spicy Tuna Roll	Roll	9.5	8	4.6
Tuna Sashimi	Sashimi	11.0	5	4.5
Mochi Ice Cream	Dessert	5.0	3	5.0

Question 1

Coleman purchased a property on Clougherty Drive, but the street number can only be determined by evaluating the following expression. What is the street number?

```
sum(2 ** np.arange(3, 7, 2)) * len(str(len("sushi" * 999)))
```

Question 2

After a few months of operation, Coleman notices that there are a few bugs in his restaurant data dashboard. For each expression below, help Coleman figure out what the expression will evaluate to, or write "error" if the expression produces an error.

Expression	Value
`menu.take(np.arange(3,0,-1)).shape[1]`	
`np.array([10, menu.get("Type").loc["Tuna Sashimi"], 1.5])[2]`	
`int(str(menu.get("Rating").max()))`	

Question 3

Ella and Bianca, a duo of world renowned pizza-turned-sushi chefs, are interviewing jointly for a job at Coleman's Catch. Help them ace their interview question by determining the output of the code block below. **Write legibly!**

```
food1 = "hamachi"
food2 = "sashimi"
food2.replace("sh", "iba").replace("s", "b") + "o".join(food1.split("a"))
```

Question 4

Ray, founder of Ray's Reviews, writes that the sushi from Coleman's Catch is delicious, but it may be overpriced. Fill in the blanks below to create a DataFrame called `money`, which is indexed by `"Type"` and has just one column, `"price_per_piece"`, that contains the highest price per individual piece of a menu item in each of the different `"Type"` categories.

```
unit_price = menu.assign(price_per_piece = __(A)__)
money = unit_price.get(__(B)__).groupby("Type").__(C)__
```

(A): _____

(B): _____

(C): _____

Question 5

Coleman has trouble selling certain menu items, so he creates the *Coleman's Catch of the Day Combo*. All menu items with ratings below 4.5 are considered clunkers. The combo consists of the most expensive clunker of each `"Type"`, which he sells as a bundle for $30.

a) Fill in the blank so that `clunkers` is a DataFrame consisting of the rows of `menu` with ratings below 4.5.

`clunkers =` _____

b) Which of the following evaluates to the amount of money a customer saves by purchasing the combo versus purchasing each item individually?

◯ `clunkers.groupby("Type").sum().get("Price").max() - 30`

◯ `30 - clunkers.groupby("Type").sum().get("Price").max()`

◯ `clunkers.groupby("Type").max().get("Price").sum() - 30`

◯ `30 - clunkers.groupby("Type").max().get("Price").sum()`

Reminders:

- **Write your TSN** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.